

Section 14.4: Tangent Planes and Linearization

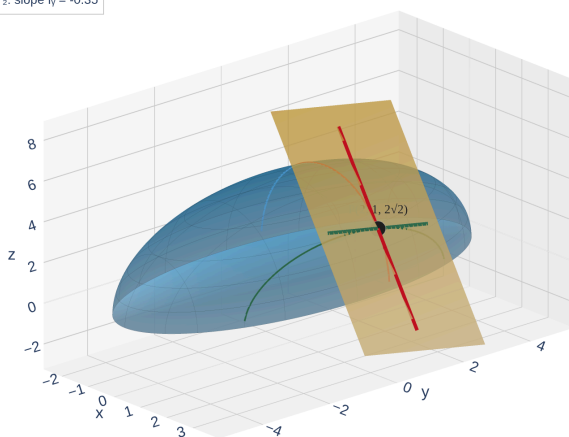
Partial Derivatives

MTH 310
Calculus III

Your Turn: Derivatives Warm-Up

From Tangent Lines to Tangent Planes

T_1 : slope $f_x = -2.83$
 T_2 : slope $f_y = -0.35$



[Interactive version](#)

Calc I: zoom in → tangent **line**. Calc III: zoom in → tangent **plane**.

The Tangent Plane Equation

Equation of the Tangent Plane

If f is differentiable at (x_0, y_0) , the tangent plane to $z = f(x, y)$ at the point (x_0, y_0, z_0) is:

$$z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

where $z_0 = f(x_0, y_0)$.

Same structure as the tangent line equation, but with **two slope terms** instead of one.

When Does the Tangent Plane Exist?

Theorem: Differentiability (Sufficient Condition)

If the partial derivatives f_x and f_y are **continuous** on an open region containing (x_0, y_0) , then f is differentiable at (x_0, y_0) .

In other words: continuous partials $\Rightarrow$ differentiable $\Rightarrow$ tangent plane exists.

Rule of thumb: If f_x and f_y are built from standard functions (polynomials, trig, exponentials, etc.) and the point is in the domain, you're safe.

Example 1: Does the Tangent Plane Exist?

Example 1

Does the hemi-ellipsoid $f(x, y) = \sqrt{25 - 4x^2 - y^2}$ have a tangent plane at $(2, 1)$? Why?

Example 2: Tangent Plane to a Hemi-Ellipsoid

Example 2

Find the equation of the tangent plane to the surface

$$z = f(x, y) = \sqrt{25 - 4x^2 - y^2}$$

at the point $(2, 1)$.

Find $f(2, 1)$, $f_x(2, 1)$, $f_y(2, 1)$, and use: $z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$

Your Turn: Find a Tangent Plane

Example 3

Find the equation of the tangent plane to

$$z = x^2 + y^2$$

at the point $(1, 2, 5)$.

Compute f_x and f_y , evaluate at $(1, 2)$, and plug into the formula.

Linear Approximation

Linearization

The **linearization** (or **linear approximation**) of f at (a, b) is:

$$L(x, y) = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$$

For (x, y) near (a, b) : $f(x, y) \approx L(x, y)$.

The tangent plane gives us a simple formula to **approximate** a complicated function near a known point.

Example 4: Approximate $f(2.05, 0.92)$

Example 4

Using the tangent plane from Example 2, approximate the value of

$$f(2.05, 0.92) = \sqrt{25 - 4(2.05)^2 - (0.92)^2}$$

Use $L(x, y) = f(2, 1) + f_x(2, 1)(x - 2) + f_y(2, 1)(y - 1)$ with $x = 2.05$, $y = 0.92$.

Differentials

Total Differential

If $z = f(x, y)$ and f is differentiable, the **total differential** is:

$$dz = f_x(x, y) dx + f_y(x, y) dy$$

where $dx = \Delta x$ and $dy = \Delta y$ are the changes in x and y .

Differentials give you the **change** in the output. Linear approximation gives you the **new value**. They're two sides of the same coin: $L = f + dz$.

Example 5: Differentials Approach

Example 5

Use differentials to approximate $f(2.05, 0.92)$ where $f(x, y) = \sqrt{25 - 4x^2 - y^2}$.
(Same problem as Example 4 — different method.)

Set $dx = 0.05$, $dy = -0.08$. Compute $dz = f_x dx + f_y dy$. Then $f(2.05, 0.92) \approx f(2, 1) + dz$.

Your Turn: Differentials Practice

Example 6

The volume of a right circular cone is $V = \frac{1}{3}\pi r^2 h$.

If the radius is measured as $r = 5$ cm and the height as $h = 12$ cm, each with a possible error of ± 0.1 cm, use differentials to estimate the maximum error in the calculated volume.

Find $dV = V_r dr + V_h dh$. Use $|dr| = |dh| = 0.1$ to get the maximum $|dV|$.

Homework

Section 14.4: 3, 9, 11, 21, 25, 27, 29, 33, 39, 43, 52